KULA

A Sharing Network for Everything

Product Specification & Development Guide

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1. Vision & Problem Statement

The Problem

Most people own far more than they use. A power drill is used for an average of 13 minutes over its entire lifetime. Children outgrow ski gear in a season. Guest rooms sit empty 90% of the year. Meanwhile, people who need these things buy them new, adding to both personal debt and ecological waste.

Existing platforms address fragments of this problem: Couchsurfing for housing, Buy Nothing groups for gifting, TaskRabbit for services, time banks for labor exchange. But no single platform integrates all of these modalities into a coherent, trust-rich social network that makes sharing as easy and natural as buying.

The Vision

Kula is a sharing network for everything. Named after the Kula ring—the ceremonial exchange system documented by Malinowski and analyzed by Marcel Mauss in The Gift—it is designed to make lending, borrowing, gifting, bartering, and time-based exchange as frictionless as possible.

The core proposition: people should be able to live significantly better lives with dramatically less dependence on monetary exchange. Kula provides the trust infrastructure, the matching intelligence, and the social fabric to make that possible.

Design Principles

- **Trust is the currency.** The entire platform is built around making trust visible, verifiable, and valuable. Every interaction either builds or erodes trust.
- **Generosity is rewarded.** Unlike extractive platforms, giving more than you receive raises your standing. The system incentivizes abundance, not scarcity.
- **Multiple exchange modes.** People choose what works for them: pure gift, temporary loan, time-dollar exchange, or direct barter. No single model is privileged.
- **Local first, universal second.** Sharing works best in communities where people can physically hand things to each other. The platform is organized around geography first, interests second.
- **Familiar interface, radical function.** The UX should feel as natural as Facebook or Instagram. The revolution is in what it enables, not in requiring users to learn new interaction patterns.
- **No surveillance architecture.** Users own their data. Verification builds trust without creating a panopticon. The platform operates in the public good, not for data extraction.

2. Core Concepts

Exchange Modes

Every listing on Kula specifies one or more exchange modes. These are the fundamental ways value moves through the network:

Mode	Description	Example
Gift	Free transfer with no expectation of return. The giver gains trust and reputation.	Surplus garden produce, outgrown children's clothes, a guest room for a visiting artist.
Loan	Temporary transfer with expected return. Both parties rate the exchange upon completion.	Power tools for a weekend project, a ski suit for a 10-year-old, camping gear for a trip.
Time Dollar	Exchange denominated in hours of labor. One hour = one time-dollar regardless of the service type.	One hour of massage exchanged for one hour of copyediting. One hour of coding for one hour of cooking lessons.
Barter	Direct trade negotiated between parties. No standardized unit of value.	Photography in exchange for Spanish tutoring. Graphic design for carpentry work.
Flexible	Poster is open to any exchange mode. Respondents can propose their preferred arrangement.	A request for help moving—could be a gift from a neighbor, a time-dollar exchange, or a barter.

Post Types

- **Offer:** I have something to share—a good, a service, a space, a skill.
- **Request:** I need something—looking for a specific item, service, or space.
- **Exchange Proposal:** I want to trade X for Y—a specific barter or time-dollar offer.

Categories

Posts are organized into the following top-level categories. Sub-categories can be added as the platform evolves:

- Housing & Space — guest rooms, studios, co-living, land access, storage
- Tools & Equipment — power tools, kitchen equipment, camping gear, electronics
- Wellness & Bodywork — massage, yoga, coaching, therapy, meditation
- Food & Garden — surplus produce, meals, seeds, composting, garden space
- Kids & Family — clothes, gear, childcare, tutoring, toys
- Creative Services — photography, design, writing, editing, music
- Transport — rides, car sharing, bikes, moving help

- Education & Skills — language, coding, cooking, crafts, mentorship
- Household — furniture, appliances, cleaning, repairs
- Professional Services — legal, accounting, consulting, admin support

3. User Types & Personas

Primary Personas

The Abundant Neighbor (Sana)

Permaculture designer in Brooklyn. Has surplus produce, herbs, and knowledge to share. Motivated by community building and ecological values. High giver-to-receiver ratio. Uses Kula primarily in Gift mode. Wants to see her neighborhood become more connected and less dependent on Amazon.

The Practical Parent (Claire)

Life coach in Park Slope with two kids. Needs to borrow seasonal items (ski gear, Halloween costumes, bikes as kids grow). Also offers coaching in time-dollars. Uses Kula across all exchange modes. Her primary motivation is practical: kids are expensive and she doesn't want to buy things she'll use once.

The Skilled Maker (Marcus)

Carpenter in Harlem with a full workshop. Loans tools regularly and occasionally barters his skills for things he needs. Motivated by craft pride and the satisfaction of sharing knowledge. Wants a system that respects the value of his tools and ensures they come back undamaged.

The Time-Rich Creative (Tomas)

Photographer in Tulum building an off-grid house. Cash-poor but skill-rich. Uses time-dollars and barter almost exclusively. Kula is how he accesses services he couldn't otherwise afford. The time-dollar system is essential to his participation.

The Community Organizer (Daniel)

Writer and public intellectual. Uses Kula both personally and as part of building community infrastructure. Posts in discussion forums, organizes local skill-shares, and advocates for the platform within his networks. Interested in Kula as a model for post-capitalist social organization.

4. Information Architecture

Navigation Structure

The primary navigation has five tabs, accessible from a persistent bottom bar (mobile) or top bar (web):

Tab	Purpose	Content
Feed	Primary content stream	Chronological/algorithmic mix of posts from your communities, filtered by type and exchange mode.
Explore	Discovery and search	Category browsing, search, trending offers/requests, nearby activity.
Groups	Community hubs	Your local and affinity groups. Community-specific feeds, member directories, and group settings.
Discuss	Moderated forums	Threaded discussions organized by community. Pinned posts, moderation tools.
Profile	Identity and reputation	Your profile, trust score, exchange history, reviews, time-dollar balance, settings.

Secondary Navigation

- **Notifications:** Accessed via bell icon in header. Responses to your posts, exchange status updates, new reviews, community announcements.
- **Messages:** Direct messaging between users, tied to specific exchanges. Accessed via envelope icon in header.
- **Create Post:** Floating action button or “+” icon. Opens structured post creation flow.
- **Settings:** Accessible from Profile tab. Account, privacy, notification preferences, verification status, time-dollar wallet.

5. User Flows

Flow 1: Onboarding

1. User downloads app or visits web app. Sees brief value proposition (3 screens max).
2. Creates account with email/phone. Sets display name (real names encouraged but not required at this stage).
3. Selects primary location (city/neighborhood). Can add additional locations later.
4. Browses suggested communities to join (minimum one required).
5. Optionally completes identity verification for “Verified” badge.
6. Optionally adds skills, bio, and profile photo.
7. Enters the feed. A welcome post from the community moderator appears at the top.

Flow 2: Creating an Offer

1. User taps “+” button. Selects “Offer” or “Request.”
2. Selects category (required) and sub-category (optional).
3. Writes title and description. Can add up to 4 photos.
4. Selects exchange mode(s): Gift, Loan, Time Dollar, Barter, or Flexible.
5. If Loan: sets expected loan duration (e.g., “1 week”).
6. If Time Dollar: sets time-dollar amount.
7. Selects which communities to post to (can be multiple).
8. Publishes. Post appears in relevant feeds.

Flow 3: Responding to a Post

1. User sees a post and taps “Respond.”
2. A message thread opens between the poster and respondent, pre-populated with the post details.
3. They negotiate specifics: pickup/delivery logistics, loan duration, barter terms, etc.
4. One party proposes a formal Exchange Agreement (a structured summary of terms).
5. Other party accepts the Agreement. Exchange status moves to “In Progress.”
6. When the exchange is complete (item returned, service rendered, etc.), both parties mark it as “Completed.”
7. Both parties are prompted to leave a mandatory rating (1–5 stars) and written review.
8. Trust scores update. Exchange is recorded in both users’ histories.

Flow 4: The Berkeley Parents Network Model

This specific flow is central to the platform’s value proposition—the ability to request something specific and have a community member provide it, potentially without the two parties ever meeting:

1. Claire posts a Request: “Looking for ski gear for a 10-year-old, just for one weekend.” Exchange mode: Flexible.

2. Marcus responds: “I have ski pants and a jacket from when my kid was that age. I can drop them off.”
3. They agree via message: Marcus will leave the gear on Claire’s porch Saturday morning. Claire will leave it back on his porch the following Sunday.
4. Exchange Agreement is created: Loan, 8 days, specific items listed.
5. Marcus drops off. Claire confirms receipt in the app.
6. Claire returns gear. Marcus confirms return.
7. Both leave reviews. Marcus’s trust score increases for generosity (he gifted delivery labor). Claire’s increases for prompt return.

Flow 5: Time-Dollar Exchange

1. Lena posts an Offer: “Massage sessions — 1 hour = 1 time-dollar.”
2. Daniel responds. He has 24 time-dollars in his wallet (earned from editing work for other members).
3. Exchange Agreement: 1 hour massage, 1 time-dollar.
4. Session happens. Both mark as complete.
5. 1 time-dollar is debited from Daniel’s wallet and credited to Lena’s.
6. Both leave reviews.

6. Data Model

The following are the core entities and their key fields. This is a logical model—specific database implementation may vary.

Users

Field	Type	Notes
id	UUID	Primary key
email	String	Unique, used for auth
display_name	String	Shown publicly
bio	Text	Optional profile bio
avatar_url	String	Profile image
primary_location	Geography	Lat/lng + city/neighborhood string
additional_locations	Array	Secondary locations (e.g., seasonal residence)
skills	Array<String>	Self-reported skills
verified	Boolean	Identity verification status

Field	Type	Notes
trust_score	Integer (0–100)	Computed from multiple signals
time_dollar_balance	Decimal	Current time-dollar wallet balance
total_exchanges	Integer	Lifetime count
total_given	Integer	Number of times user was the giver/provider
total_received	Integer	Number of times user was the receiver
created_at	Timestamp	Account creation date
last_active	Timestamp	For activity tracking

Posts

Field	Type	Notes
id	UUID	Primary key
author_id	UUID (FK)	References Users
type	Enum	offer | request | exchange_proposal
exchange_modes	Array<Enum>	gift | loan | time_dollar | barter | flexible
category	String	Top-level category
subcategory	String	Optional
title	String	Max 120 characters
body	Text	Full description
images	Array<URL>	Up to 4 images
loan_duration	String	If loan mode: expected duration
time_dollar_amount	Decimal	If time-dollar mode: price
communities	Array<UUID>	Which communities it's posted to
status	Enum	active | fulfilled | expired | closed
created_at	Timestamp	

Exchange Agreements

Field	Type	Notes
id	UUID	Primary key
post_id	UUID (FK)	The originating post
provider_id	UUID (FK)	The person giving/lending/serving

Field	Type	Notes
receiver_id	UUID (FK)	The person receiving
exchange_mode	Enum	The agreed-upon mode for this exchange
terms	Text	Free-text description of what was agreed
time_dollar_amount	Decimal	If applicable
loan_return_date	Date	If loan: expected return
status	Enum	proposed | accepted | in_progress | completed | disputed | cancelled
provider_confirmed	Boolean	Provider marked as complete
receiver_confirmed	Boolean	Receiver marked as complete
created_at	Timestamp	
completed_at	Timestamp	

Reviews

Field	Type	Notes
id	UUID	Primary key
exchange_id	UUID (FK)	References Exchange Agreements
author_id	UUID (FK)	Who wrote the review
subject_id	UUID (FK)	Who is being reviewed
rating	Integer (1–5)	Star rating
body	Text	Written review (required, min 20 characters)
created_at	Timestamp	

Communities

Field	Type	Notes
id	UUID	Primary key
name	String	e.g., “NYC — Manhattan”
type	Enum	geographic | affinity
description	Text	Community description
location	Geography	Center point + radius (geographic communities)
member_count	Integer	

Field	Type	Notes
moderators	Array<UUID>	User IDs with moderator privileges
settings	JSON	Posting rules, approval requirements, etc.
created_at	Timestamp	

Additional Entities

- **Messages:** Direct messages between users, linked to Exchange Agreements. Standard message table with sender, receiver, body, timestamp, read status.
- **Discussion Threads:** Forum-style posts within communities. Thread table with title, body, author, community, pinned status. Replies as a separate table with parent_thread_id.
- **Notifications:** System-generated notifications for exchange events, new responses, reviews, community announcements. Standard notification table with type, recipient, content, read status.
- **Time-Dollar Ledger:** Double-entry bookkeeping. Every time-dollar transaction creates two entries: a debit from one user and a credit to another, linked to an Exchange Agreement.
- **Reports:** User reports for moderation. Reporter, reported user/post, reason, status, moderator notes.

7. Trust & Reputation System

The trust system is the backbone of Kula. Without trust, sharing between strangers doesn't work. The system must be transparent enough that users understand how to build trust, but sophisticated enough to resist gaming.

Trust Score Computation

The trust score is a composite number from 0 to 100, computed from the following weighted signals:

Signal	Weight	Description
Review average	25%	Average star rating from all reviews received. Minimum 3 reviews before this signal is active.
Exchange volume	15%	Total number of completed exchanges. Logarithmic scaling (diminishing returns after ~50 exchanges).
Generosity ratio	20%	Ratio of given-to-received. A 2:1 ratio earns full marks. This is the key differentiator from commercial platforms.

Signal	Weight	Description
Consistency	15%	Standard deviation of ratings. Low variance = reliable = higher score.
Verification	10%	Identity verification adds a fixed bonus. Additional bonuses for community endorsements.
Response rate	10%	Percentage of messages responded to within 24 hours.
Activity recency	5%	Decay function: trust slowly decreases if user is inactive for 60+ days. Prevents ghost accounts from maintaining high scores.

Trust Tiers

- **Highly Trusted (85–100):** Displayed with a green shield badge. Can post in all communities, propose exchanges without pre-approval, and access premium features like hosting.
- **Established (60–84):** Displayed with a gold shield badge. Standard access to all features.
- **Building (0–59):** Displayed with an amber badge. May require moderator approval for certain post types in some communities. New users start at 30 and build from there.

Anti-Gaming Measures

- Reviews are mandatory for continued platform use. After 3 completed exchanges without reviews, posting privileges are suspended until reviews are submitted.
- Reciprocal review detection: if two users only review each other, those reviews are down-weighted.
- New accounts cannot leave reviews above 4 stars for their first 5 exchanges (prevents shill account creation).
- Trust score changes are rate-limited: no more than +/- 5 points per day to prevent sudden manipulation.
- Community moderators can flag suspicious patterns for admin review.

8. Exchange Mechanics

Exchange Lifecycle

Every exchange follows a structured lifecycle with clear status transitions:

Status	Triggered By	What Happens
Proposed	Respondent sends a response to a post	A message thread opens. Parties negotiate details.
Accepted	Both parties agree to terms via Exchange Agreement	Terms are formalized. For time-dollar exchanges, funds are held in escrow.
In Progress	Provider confirms handoff (or service begins)	Exchange is live. Countdown begins for loans.
Completed	Both parties mark as complete	Review prompts are sent. Time-dollars are transferred. Stats update.
Disputed	Either party raises a concern	Moderator is notified. Exchange is paused pending resolution.
Cancelled	Either party cancels before In Progress	No impact on trust scores. Escrowed time-dollars returned.

Loan-Specific Mechanics

- Loans have an explicit return date set in the Exchange Agreement.
- Automated reminders are sent 2 days before, on the day of, and 1 day after the return date.
- Late returns (3+ days) trigger a notification to both parties and a flag on the exchange record.
- Persistent late returns (7+ days) negatively impact the borrower's trust score.
- The lender can mark an item as "not returned" which escalates to moderator review.

Condition Reporting

For loans of physical items, both parties can document condition:

- Provider takes optional photos before handoff, tagged to the exchange.
- Receiver takes optional photos upon receipt.
- Upon return, provider can flag any damage with photos.
- Damage claims are handled through the dispute resolution flow.

9. Time-Dollar Economy

The time-dollar system is inspired by Time Republik and traditional time banking. The fundamental principle: one hour of anyone's time is worth one time-dollar, regardless of the service. A plumber's hour equals a tutor's hour equals a massage therapist's hour. This is a deliberate philosophical choice that challenges market-based valuation of labor.

How It Works

- **Earning:** When you provide a service denominated in time-dollars, the receiver's wallet is debited and yours is credited.
- **Spending:** When you receive a service, your wallet is debited.
- **Starting balance:** New users receive 5 time-dollars upon account creation and identity verification. This prevents a cold-start problem where new users can't access the system.
- **Negative balances:** Users can go up to -3 time-dollars to prevent the system from being too rigid. Beyond -3, they must earn before spending.
- **Fractional amounts:** Time-dollars can be denominated in 15-minute increments (0.25 TD). A 30-minute tutoring session costs 0.5 TD.

Safeguards

- Time-dollars cannot be purchased with money. They can only be earned through service.
- Time-dollars cannot be transferred without an associated Exchange Agreement. No speculative hoarding.
- Maximum balance cap of 100 time-dollars per user to prevent accumulation without giving back.
- Time-dollars expire after 12 months of account inactivity.

Future: Convertibility

In a future phase, the platform may explore limited convertibility between time-dollars and other value systems (community currencies, crypto tokens, or even fiat). This is deliberately excluded from the MVP to keep the system simple and to preserve the philosophical integrity of the time-dollar as a non-market unit of value. Any future convertibility would need extensive community discussion.

10. Kula Circles: AI-Powered Multi-Party Matching

Kula Circles is the platform's most transformative feature: an AI-powered matching engine that discovers multi-party exchange cycles, solving the fundamental problem that has historically limited every non-monetary exchange system.

The Problem: Double Coincidence of Wants

In any non-monetary exchange system, a transaction requires that two people each have what the other wants at the same time. Economists call this the "double coincidence of wants." It is the reason money was invented and the reason most time banks and barter networks plateau at a few hundred users. The matching surface is simply too thin for one-to-one exchanges to reliably serve a diverse community.

Daniel wants a household organizer. He can offer writing and editing. But the household organizer he finds doesn't need editing—she needs someone to drive her kid to soccer practice. In a money economy, this is trivially solved: Daniel pays cash, she pays a babysitter cash. But without money as the intermediary, the exchange is impossible—unless the system can find a chain.

The Solution: Exchange Cycles

Kula Circles discovers multi-party exchange cycles where each participant gives what they can and receives what they need, even though no two participants exchange directly with each other. A circle might look like this:

1. Daniel offers 3 hours of manuscript editing to Sofia (a writer who posted a request for editorial help).
2. Sofia offers 3 hours of graphic design to Marcus (a carpenter who needs a logo for his furniture business).
3. Marcus offers to drive Lena's daughter to soccer practice twice a week for a month (a task Lena posted as a need).
4. Lena, now freed up on Tuesday afternoons, offers 3 hours of household organizing to Daniel.

Nobody in that chain exchanged directly with the person they received value from. But everyone got what they needed and gave what they could. The AI found a four-person cycle that no individual could have discovered on their own.

Precedent: kidney exchange programs. This is not a theoretical concept. Paired kidney exchange programs use the same algorithmic approach to find cycles among incompatible donor-patient pairs. Chains of three, four, or five links are routine in transplant matching. What Kula Circles does is apply the same graph-cycle algorithms to the full range of goods, services, and skills.

How It Works: Technical Overview

1. The Offer/Need Layer

Every user maintains two living profiles in addition to their standard posts:

- **What I can offer:** A list of skills, goods, and services the user is willing to provide. These can be entered explicitly ("I can edit manuscripts, facilitate workshops, and cook for groups") or inferred by the AI from the user's profile, past exchanges, and post history.
- **What I need:** A list of things the user would value receiving. These can be specific ("I need someone to organize my apartment") or general ("I'm looking for bodywork, childcare help, or someone to teach me Spanish").

Crucially, these lists are maintained as natural-language descriptions, not rigid categories. The AI uses semantic understanding to match "I need help getting my house in order" with "professional organizing and decluttering"—something a keyword-based system would miss.

2. The Exchange Graph

The system constructs a directed graph where:

- Each node represents a user.
- A directed edge from User A to User B exists when User A can offer something User B needs.
- Each edge carries weights: semantic match quality (how well the offer meets the need), geographic proximity, schedule compatibility, trust compatibility, and exchange mode preference alignment.

This graph is updated continuously as users modify their profiles, complete exchanges, and post new offers or requests.

3. The Cycle-Finding Algorithm

The core engine searches the graph for cycles—closed loops where every participant both gives and receives. Key parameters:

- **Cycle length:** Minimum 2 (a direct match), maximum 6. Longer chains become logistically fragile and hard for users to understand. The sweet spot is 3–4 participants.
- **Cycle scoring:** Each discovered cycle is ranked by: total match quality (sum of edge weights across the cycle), minimum trust score in the chain (the weakest link determines overall reliability), geographic feasibility (are participants close enough to exchange?), and time alignment (can all exchanges happen within a reasonable window?).
- **Optimization:** The system doesn't just find any cycle—it finds the set of non-overlapping cycles that maximizes total satisfaction across the network. This is a variant of the maximum weight cycle cover problem, which is NP-hard in the general case but tractable with heuristic approaches at the scale of a community network (hundreds to low thousands of users).

4. Semantic Matching via AI

This is where the LLM layer makes the system fundamentally different from anything that could have been built five years ago. The AI performs several matching functions:

- **Need/offer alignment:** Understanding that "I need help with my garden" matches both "permaculture design consultation" and "weeding and planting labor"—but that the first is a skill match and the second is a time match, and they should be valued differently.
- **Implicit skill inference:** A user who has written three books and completed 15 editing exchanges doesn't need to explicitly list "editing" as a skill. The system knows.
- **Reciprocity analysis:** Understanding that a 1-hour massage and 1 hour of coding help are both "one hour" in time-dollar terms, but that the massage therapist might prefer to receive something other than time-dollars—perhaps childcare or a home-cooked meal. The AI can factor in preference patterns when proposing circles.
- **Natural-language proposal generation:** Rather than presenting users with an abstract graph diagram, the AI generates a clear, human-readable explanation of each proposed circle: who gives what to whom, why the system thinks this match works, and what each participant gets out of it.

The Proposal Interface

When the system discovers a viable Kula Circle, it presents it to all participants simultaneously via a dedicated notification. The proposal includes:

- A visual circle diagram showing all participants with their avatars, trust scores, and roles.
- A natural-language summary: “We found a 4-person exchange circle that gets you the household organizing help you’ve been looking for.”
- Each participant’s specific role: what they give, what they receive, and to/from whom.
- Profiles and reviews of every participant in the circle, so trust can be evaluated.
- A proposed timeline for all exchanges in the circle.
- Accept / Decline / Suggest Modification buttons.

Activation: A circle only activates when every participant accepts. If one person declines, the system searches for alternative cycles that exclude that person. Participants can also suggest modifications (“I can do this, but next week instead of this week”) which are relayed to the group.

Handling Incomplete Circles

The risk of multi-party exchanges is that one person fails to follow through, breaking the chain. Kula Circles addresses this through several mechanisms:

- **Sequential confirmation:** Each exchange in the circle is confirmed as complete before the next one is expected to begin. If Daniel finishes editing Sofia’s manuscript, that’s confirmed before Marcus is expected to begin his contribution.
- **Trust gating:** Circles are only proposed to users with a minimum trust score of 50. Higher-value or longer circles may require higher minimums.
- **Fallback to time-dollars:** If one link in a circle breaks (someone doesn’t follow through), the affected participant receives time-dollar compensation equivalent to the value they provided. This is funded from a platform-level time-dollar reserve or debited from the non-compliant user’s account.
- **Reputation impact:** Breaking a circle has a more significant trust score penalty than failing on a one-to-one exchange, because it affects multiple people.

Example Scenarios

Scenario 1: The Skill Circle

Daniel (editing) → Sofia (graphic design) → Marcus (carpentry) → Lena (organizing) → Daniel. Four professionals, each working in their zone of competence, each receiving something they need. Total time: roughly 3 hours each. No money changes hands. The AI matched them based on posted needs and inferred skills.

Scenario 2: The Parent Chain

Claire needs ski gear for her daughter. Raj has ski gear but needs someone to tutor his son in math. Elena can tutor math but needs a ride to work next week while her car is in the shop. Tom drives past Elena’s office daily and his daughter outgrew her ski gear—wait, that’s a different set. The AI finds that Claire can help Raj’s son with reading (she’s a life coach who used to teach), Raj lends the ski gear to Claire, and the circle closes

through a third link. The AI discovered that Claire's coaching background included childhood literacy—a match no human would have thought to propose.

Scenario 3: The Cross-Community Circle

A circle that spans the NYC and Tulum communities: Tomas (Tulum photographer) shoots portfolio images for a NYC-based writer's website, the writer edits a grant proposal for a Brooklyn permaculture designer, and the designer provides a remote consultation on Tomas's food forest project in Tulum. Geography is no barrier for services that can be delivered digitally.

Data Model Additions

The following additions to the data model are required to support Kula Circles:

Entity / Field	Type	Notes
User: offers_list	Array<Text>	Natural-language descriptions of what the user can offer. Supplemented by AI inference from profile and history.
User: needs_list	Array<Text>	Natural-language descriptions of what the user currently needs.
User: circle_preferences	JSON	Preferred circle size, preferred exchange modes, schedule availability, maximum commitment hours.
KulaCircle	New Entity	Represents a proposed or active multi-party exchange cycle.
KulaCircle: id	UUID	Primary key.
KulaCircle: status	Enum	proposed | active | completed | broken | expired
KulaCircle: participants	Array<UUID>	Ordered list of user IDs forming the circle.
KulaCircle: exchanges	Array<UUID>	References to Exchange Agreements for each link in the chain.
KulaCircle: match_score	Float	The algorithm's confidence score for this circle.
KulaCircle: proposed_at	Timestamp	When the circle was proposed to participants.
KulaCircle: activated_at	Timestamp	When all participants accepted.
KulaCircle: completed_at	Timestamp	When all exchanges in the circle are confirmed complete.
KulaCircle: ai_summary	Text	AI-generated natural-language description of the circle.

Implementation Phasing

Kula Circles is not an MVP feature. It requires a critical mass of users with populated offer/need profiles before the cycle-finding algorithm has enough data to work. The recommended rollout:

- **Phase 1 (MVP):** Collect offers_list and needs_list data from users during onboarding and profile editing. Store this data even though the matching engine doesn't exist yet. This ensures the graph is populated when the engine launches.
- **Phase 2:** Launch "Smart Matches"—simple AI-powered one-to-one suggestions. "You need editing help. Sana in your community has asked for someone to help plan an event, and you've facilitated events before. Would you like to propose an exchange?" This builds user comfort with AI-mediated matching.
- **Phase 3:** Launch full Kula Circles with multi-party matching. Start with circles of 3 (simplest), expand to 4–6 as the system proves reliable.
- **Phase 4:** Advanced features: real-time circle discovery (notifying users the moment a viable circle forms), standing circles (recurring exchange arrangements), and cross-community circles.

Why This Changes Everything

Kula Circles does something that money actually cannot do as well: it finds exchanges where the value is incommensurable in market terms but perfectly complementary in human terms. Three hours of Daniel's editing is "worth" $300 on the open market. Three hours of household organizing is "worth" $120. In a money economy, that asymmetry creates a barrier—Daniel would have to pay the difference, or the organizer would feel shortchanged. In a Kula Circle denominated in time, that asymmetry is irrelevant. Both gave three hours. Both got what they needed. The market's valuation of their labor is simply not part of the equation.

This is the philosophical heart of the platform: a computational system that enables genuinely post-market exchange at scale, not by abolishing the concept of value but by recognizing forms of value that markets systematically fail to capture.

11. Communities & Local Groups

Community Types

- **Geographic communities:** Defined by a location and radius. Examples: "NYC — Manhattan," "Westchester," "Tulum." Users are automatically suggested geographic communities based on their location. These are the primary organizational unit of the platform.
- **Affinity communities:** Interest-based or profession-based groups that cross geographic boundaries. Examples: "Writers & Editors Guild," "Parents Network," "Permaculture Circle." These are secondary and supplement geographic communities.

Community Features

- Community-specific feed showing only posts from that group.

- Member directory with trust scores and skills.
- Community discussion board (see Section 12).
- Community-elected moderators (see Section 12).
- Community statistics: total exchanges, active members, most-shared categories.
- Community announcements (pinned posts from moderators).
- Option for communities to require moderator approval for new posts (configurable per community).

Creating Communities

Any user with a trust score of 60+ can propose a new community. Geographic communities require a minimum of 10 interested members before activation. Affinity communities require 5. Community proposals are reviewed by platform admins to prevent fragmentation and duplication.

12. Discussion & Moderation

Discussion Board

Each community has a threaded discussion board for conversations that aren't exchange-specific. Examples: proposals for local sharing events, debates about community guidelines, coordination of group activities, questions and introductions.

- Threads have a title, body, and optional tags.
- Replies are threaded (one level deep, not infinitely nested).
- Moderators can pin, lock, or archive threads.
- Users can flag threads or replies for moderator review.

Moderation System

- **Community moderators:** Elected by community members via a simple vote, or appointed by community founders. Minimum 2 moderators per community. Moderators can approve/reject posts, remove content, issue warnings, and temporarily suspend users from their community.
- **Platform moderators:** Employed or appointed by the Kula team. Handle cross-community issues, escalated disputes, identity verification, and platform-wide policy enforcement.
- **Reporting flow:** Any user can report a post, message, or user. Reports go to community moderators first. If unresolved within 48 hours or if the report concerns a moderator, it escalates to platform moderators.
- **Escalation tiers:** Warning → temporary suspension from community → permanent ban from community → platform-wide suspension.

13. Notifications & Messaging

Notification Types

- Someone responded to your post.
- Exchange Agreement proposed / accepted / completed.
- New review received.
- Loan return date approaching / overdue.
- New post in your community matching your interests (optional, can be configured).
- Community announcement from moderator.
- Trust score milestone reached.
- Time-dollar wallet credited / debited.

Messaging

Direct messaging is exchange-centric: every conversation is linked to a post or Exchange Agreement. This prevents the platform from becoming a general-purpose chat app (which dilutes its purpose) and provides context for every conversation. Users cannot message strangers without first responding to a post or being in the same community.

14. Search & Discovery

- **Full-text search:** Search across post titles, descriptions, user names, user skills, and community names.
- **Filters:** Post type (offer/request), exchange mode, category, distance, trust score minimum, availability window.
- **Nearby:** Map view showing active offers/requests within a configurable radius.
- **Recommendations:** Based on your past exchanges, skills, and community membership. “People who borrowed camping gear also borrowed...”
- **Alerts:** Users can set saved searches with notifications. “Notify me when someone in Westchester offers yoga instruction.”

15. Identity & Verification

Verification Tiers

- **Basic (required):** Email verification + phone number. Minimum for account creation.
- **Verified (optional, strongly encouraged):** Government ID check via a third-party service (e.g., Stripe Identity, Jumio). Adds the green verified badge. Grants +10 trust score bonus and access to housing/high-value exchanges.

- **Community Vouched:** Existing members with 80+ trust scores can vouch for new members they know personally. Three vouches from established members provide an alternative path to Verified status without government ID.

Privacy Principles

- Users control what information is visible on their profile.
- Location is shown at the neighborhood level, never exact address.
- Verification status is visible but the underlying documents are never stored or shared.
- Users can export or delete all their data at any time.
- The platform does not sell, share, or monetize user data.

16. Technical Architecture

Recommended Stack

Layer	Technology	Rationale
Frontend (Mobile)	React Native or Expo	Cross-platform (iOS + Android) from a single codebase. Largest developer pool. Fast iteration.
Frontend (Web)	Next.js (React)	SEO-friendly, server-side rendering, shared component library with mobile.
Backend	Node.js (Express or Fastify) or Python (FastAPI)	Rapid development, large ecosystem. Python preferred if ML/recommendation engine planned.
Database	PostgreSQL	Mature, reliable, excellent geospatial support (PostGIS), strong JSON support for flexible schemas.
Auth	Supabase Auth or Auth0	Handles email/phone/social login, MFA, and session management.
Real-time	Supabase Realtime or Socket.io	For messaging and live notifications.
File Storage	S3-compatible (AWS S3 or Supabase Storage)	Profile images, post images, verification documents.
Search	PostgreSQL full-text search (MVP) → Meilisearch or Typesense (scale)	Start simple, upgrade when query complexity warrants it.
Geolocation	PostGIS + Mapbox or Google Maps	For nearby discovery and community boundaries.
Identity Verification	Stripe Identity or Jumio	Third-party service for government ID checks.
Hosting	Vercel (web) + Railway or Render (backend) or AWS	Managed services reduce ops burden for small team.

Layer	Technology	Rationale
AI / Matching	Anthropic Claude API (semantic matching) + custom graph algorithms	LLM for semantic need/offer matching and natural-language circle proposals. Graph-cycle algorithms for Kula Circles.

Architecture Principles

- API-first design: the backend exposes a REST or GraphQL API consumed by both mobile and web clients.
- Start monolithic, decompose later. Microservices are premature for an MVP.
- Comprehensive logging from day one. Every exchange, trust score change, and moderation action is logged.
- Rate limiting on all public endpoints. Abuse prevention from the start.
- Automated backups of the database. Non-negotiable.

17. MVP Scope & Phasing

Phase 1: Core MVP (Months 1–3)

Goal: a working sharing network for a single community (one neighborhood) with 50–100 users.

Includes:

- User registration and basic profile (email + phone auth).
- Post creation: offers and requests with categories and exchange modes.
- Feed with chronological display and basic filters.
- Direct messaging tied to posts.
- Simple Exchange Agreement flow (propose → accept → complete).
- Mandatory review system after each exchange.
- Basic trust score (simplified computation: review average + exchange count).
- One geographic community with a discussion board.
- Offers and needs profile fields collected during onboarding (populating the graph for future Kula Circles).
- Web app only (responsive, mobile-friendly).

Excludes from Phase 1:

- Native mobile apps.
- Time-dollar system.
- Identity verification beyond email/phone.
- Geolocation features (map view, nearby discovery).
- Saved searches and alerts.
- Community creation by users.
- Advanced moderation tools.
- Kula Circles (AI-powered multi-party matching).

Phase 2: Community Growth (Months 4–6)

- Time-dollar wallet and exchange flow.
- Identity verification (Stripe Identity integration).
- Multiple communities (geographic + affinity).
- Community moderator tools (approve/reject, pin, suspend).
- Notification system (in-app + email).
- Improved search with filters.
- Loan lifecycle (return dates, reminders, overdue tracking).
- Full trust score algorithm with all weighted signals.
- Smart Matches: AI-powered one-to-one exchange suggestions based on offer/need profiles.

Phase 3: Scale (Months 7–12)

- Native mobile apps (React Native / Expo).
- Map-based discovery of nearby offers/requests.
- Recommendation engine.
- Kula Circles: full multi-party exchange cycle matching (3–6 person chains).
- Saved searches with alerts.
- Community creation by users.
- Advanced analytics dashboard for community moderators.
- Onboarding refinement based on user research.
- Internationalization (multi-language support).

Phase 4: Ecosystem (Year 2+)

- API for third-party integrations.
- Potential token or credit system (research and community governance required).
- Federation with other sharing networks.
- Institutional accounts (co-ops, community organizations, tool libraries).
- Impact reporting (carbon saved, money saved, community connections made).

- Advanced Kula Circles: standing circles (recurring exchanges), real-time circle discovery, cross-community circles.

18. Metrics & Success Criteria

MVP Success (Phase 1)

- 50+ active users in the pilot community.
- 100+ completed exchanges in the first 3 months.
- Average review score above 4.0.
- At least 30% of users have both given and received (indicating reciprocal participation).
- Weekly active user rate above 40%.

Growth Metrics

- **North Star Metric:** Completed exchanges per week per community. This captures both engagement and value delivery.
- **Secondary metrics:** New user registrations, user retention (7-day, 30-day), time to first exchange, average trust score across the network, time-dollar velocity (how fast they circulate), community-to-community spread.
- **Kula Circles metrics (Phase 3+):** Circles proposed per week, circle acceptance rate, circle completion rate, average circle size, percentage of exchanges occurring via circles vs. direct, user satisfaction with circle matches.
- **Health metrics:** Dispute rate (should be below 2%), average response time, review completion rate (target: 90%+), moderator response time.

19. Competitive Landscape

Platform	What It Does	Gap Kula Fills
Couchsurfing	Free housing exchange for travelers.	Kula extends the model beyond housing to all goods and services. CS has also shifted to a paid model and lost community trust.
Buy Nothing	Hyperlocal gifting groups on Facebook.	Kula adds structure (trust scores, exchange agreements, multiple modes), moves off Facebook's infrastructure, and adds time-dollar and barter options.

Platform	What It Does	Gap Kula Fills
Time Republik	Global time banking platform.	Kula integrates time banking with gifting, loaning, and bartering in a single social network rather than siloing it.
Freecycle / Craigslist Free	Free stuff listings.	Kula adds trust, community, and exchange modes beyond one-way gifting.
TaskRabbit / Fiverr	Paid service marketplace.	Kula enables the same service exchange without money via time-dollars and barter.
Nextdoor	Neighborhood social network.	Kula is purpose-built for sharing and exchange, not general neighborhood chatter. Deeper trust and exchange mechanics.
OLIO	Food sharing app.	Kula covers all categories, not just food. Broader exchange modes.

Kula's unique position: No existing platform combines all five exchange modes (gift, loan, time-dollar, barter, flexible) with a robust trust system, local community structure, moderated discussion, and AI-powered multi-party matching (Kula Circles)—all in a single, familiar social network interface. The Kula Circles feature in particular has no equivalent in any existing sharing platform and represents a genuine technical and social innovation.

20. Open Questions

The following questions require decisions before or during development. They are listed here to ensure they are not overlooked:

8. **Revenue model:** How does Kula sustain itself? Options include: freemium membership tiers, optional verified-identity fees, community-level subscriptions for organizations, grants and philanthropic funding, or a small platform fee on time-dollar transactions. Each has different implications for the platform's values.
9. **Real names vs. pseudonyms:** Should real names be required? Real names increase trust but reduce privacy. Pseudonyms with verification offer a middle path. The current spec encourages but does not require real names.
10. **Governance:** As the platform grows, who makes decisions about policy, features, and moderation standards? A cooperative governance model (one user, one vote) aligns with the platform's values but is complex to implement. Advisory board? DAO-like structure? Community council?

11. **Geographic expansion strategy:** Launch in one neighborhood and expand organically? Seed multiple cities simultaneously? Target existing communities (co-ops, mutual aid networks, parent groups) for adoption?
12. **Insurance and liability:** If someone damages a loaned item, what recourse exists beyond trust score penalties? If someone is injured during a bartered service? The platform likely needs a Terms of Service that clarifies Kula is a facilitator, not a guarantor.
13. **Accessibility:** How to serve users without smartphones or reliable internet? Shared community devices? SMS-based participation for core features?
14. **Time-dollar philosophy:** Is radical equality (one hour = one hour regardless of service) the right choice? Some time banks allow negotiated rates. This is a values decision, not a technical one.
15. **Data portability and federation:** Should users be able to export their trust history to other platforms? Should Kula interoperate with other sharing networks via an open protocol?

End of Specification

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